

Unit 6b: Atmospheric Heat, Energy, Motion & Global Circulation

Reading: pages 83–94 · 20 points · 15 minutes · Closed book

Name _____

Date _____

Score / 20

Answer every question. For multiple choice, circle the single best answer. Write short answers in complete sentences.

Part A · Multiple Choice

6 questions × 2 points = 12 points

1. Albedo is a measure of
 - a) how fast the atoms in a material vibrate
 - b) the amount of water vapor in the air
 - c) how well a surface reflects light
 - d) the air pressure at a given altitude
2. *Latent heat* is the heat taken in or released when a substance
 - a) is compressed
 - b) reflects sunlight
 - c) rises in temperature
 - d) changes state
3. Water has a very high specific heat. This means that
 - a) it takes a great deal of energy to change the temperature of water
 - b) water reflects almost all the light that strikes it
 - c) water cannot change state
 - d) water heats up faster than asphalt in sunlight
4. In the three-cell model, the Hadley cell is the convection cell found between
 - a) 0° and 30° latitude
 - b) 30° and 60° latitude
 - c) 60° latitude and the pole
 - d) the two poles only
5. Surface winds in the Hadley cell of the Northern Hemisphere blow from the northeast toward the southwest. These are called
 - a) the westerlies
 - b) the trade winds
 - c) the polar easterlies
 - d) the jet stream

6. Chinook winds are warm and dry on the leeward side of a mountain range because
- a) the leeward side receives more direct sunlight
 - b) air that rose, cooled, and dropped its precipitation on the windward side then descends and warms
 - c) cold ocean air is drawn inland over the range
 - d) the mountains block the jet stream

Part B • True or False

2 questions × 1 point = 2 points

7. Wind belts are named for the direction the winds are blowing toward.
TRUE / FALSE
8. A jet stream forms where there is a large temperature difference between two air masses.
TRUE / FALSE

Part C • Short Answer

2 questions × 3 points = 6 points

9. Explain the difference between *heat* and *temperature*. Then say which has the higher temperature and which contains more heat: a candle flame, or a bathtub full of hot water.

10. Name the three convection cells of the three-cell model in one hemisphere, and give the approximate latitude range of each.

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